

ABANGCEBU AI — SPRINT 1 SYSTEM ARCHITECTURE

Landing Page Wireframe & Layout Specification

Google Maps-Inspired Full-Bleed Map Canvas Architecture for Desktop (1440px) & Mobile (100dvh)

Jira Reference: SCRUM-64

Framework: Next.js 16 + React 19 + MapLibre GL JS v6

Status: APPROVED SPECIFICATION

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Sprint: Sprint 1 (Architecture & Governance)

GitHub Source: github.com/hermarDev/Abang_Cebu_AI_Group2/blob/main/docs/design/landing-page-wireframe.md

1. Executive Summary & Design Philosophy

AbangCebu AI replaces traditional static classifieds with an authentic **map-first discovery interface** inspired directly by Google Maps. In this architecture, **the landing page IS the map**—finding rental accommodations across Metro Cebu (CIT-U, USC, IT Park, Urgello, Banilad) begins immediately on an interactive vector canvas without forced registration paywalls.

Core Product Axiom: "Browse First, Authenticate When Value is Clear." Seekers can pan, zoom, filter, and inspect verified properties immediately. Authentication is strictly deferred until high-intent actions (saving favorites, contacting verified landlords, or booking oculars).

2. Desktop Viewport Specification (1440px × 900px)

The desktop viewport renders a **100vw × 100vh full-bleed MapLibre canvas**. Overlaid on the left is a floating search & results card (400px width), collapsible via a chevron toggle. Map centering accounts for the card's width via optical viewport offset padding (`left: 432px`) so pins are never obscured.

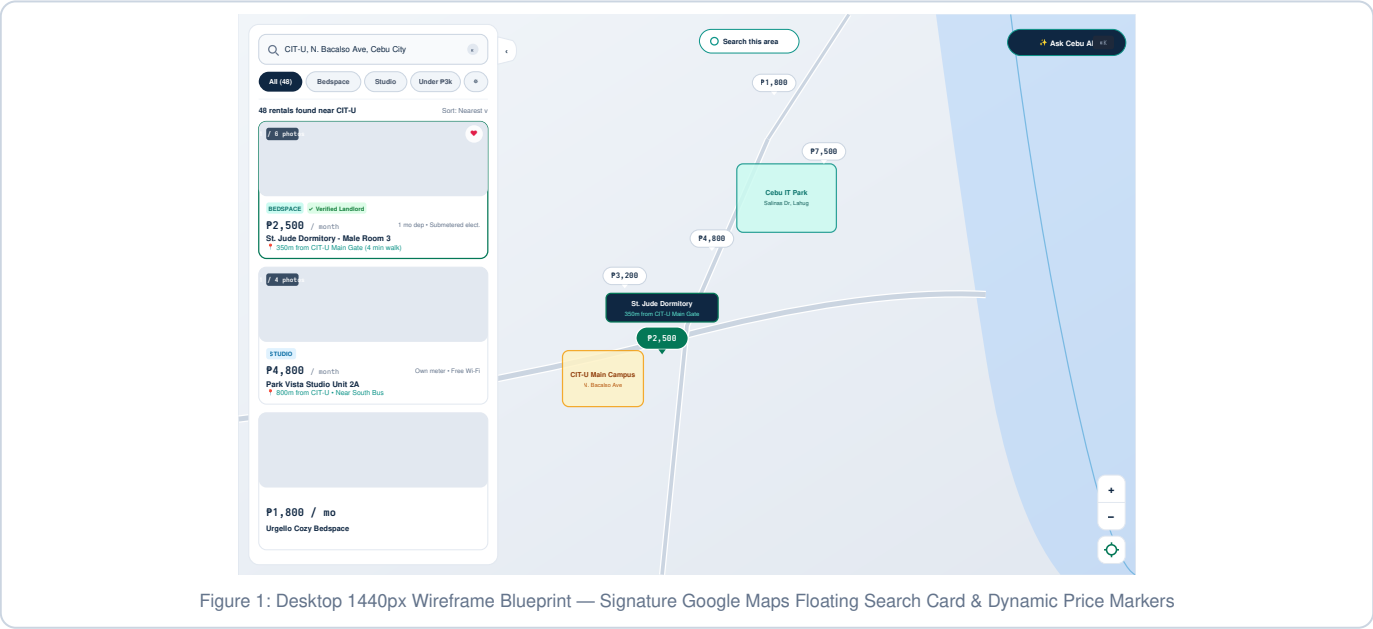


Figure 1: Desktop 1440px Wireframe Blueprint — Signature Google Maps Floating Search Card & Dynamic Price Markers

Component	Dimensions / Placement	Interaction & Ergonomics
Floating Left Card	<code>w-[400px] h-[calc(100dvh-32px)]</code>	Glassmorphic card (<code>bg-card/95 backdrop-blur-md shadow-2xl</code>). Houses search input, category pills, and scrollable rental feed.
Collapsible Toggle	<code>w-9 h-11</code> (chevron button)	Slides panel off-screen (<code>-translate-x-[calc(100%+24px)]</code>) for 100% full-screen map exploration. Re-opens via edge tab.
Dynamic Price Markers	HTML DOM Price Badges	Default white pill (<code>P2,500</code>); Hovered navy pill (<code>scale-110</code>); Active emerald pin (<code>scale-115 ring-4 ring-emerald-200</code>).
Map Overlay HUD	Top-Center, Bottom-Right	Floating "Search this area" button (renders on pan >500m), zoom stack (+/-), GPS Locate Me, and "Ask Cebu AI" FAB.

3. Mobile Viewport Specification (375px – 412px, 100dvh)

The mobile viewport is standardized on **100dvh** (Dynamic Viewport Height) to eliminate iOS Safari address bar clipping. Overlaid on the full-bleed map are a top floating search pill and a **3-Snap Point Bottom Sheet Drawer**.

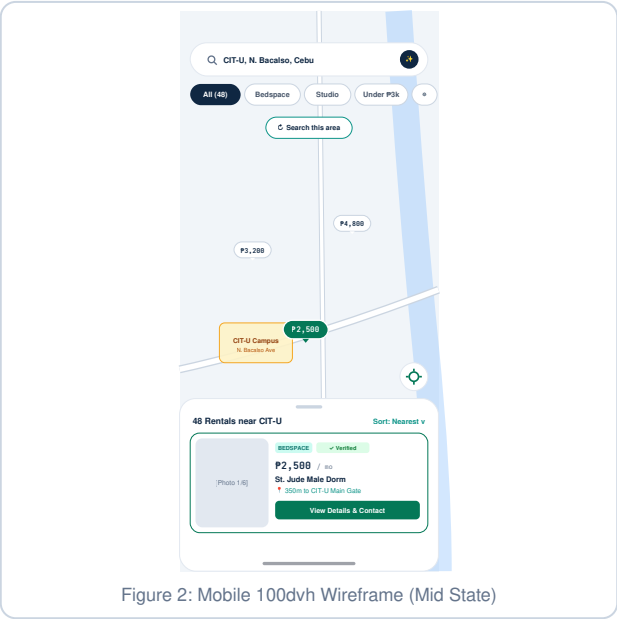


Figure 2: Mobile 100dvh Wireframe (Mid State)

Mobile 3-Snap Bottom Sheet Mechanics:

Snap State	Height	User Experience & Visible Feed
Peek State	88px (15dvh)	Map is ~85% visible. Shows drag handle, "14 rentals found" summary, and quick sort pill.
Mid State	48dvh (~400px)	Split mode: Top 52% map active. Bottom sheet shows horizontal carousel of listing cards (snap-x).
Full State	88dvh	High-density listing feed. Full vertical scrollable card list for focused property comparison.

Mobile Touch Ergonomics & Guardrails:

- **44x44px Minimum Touch Targets:** Enforced via `@utility touch-target` for all chips, buttons, and close triggers.
- **Anti-Zoom 16px Rule:** Search inputs set to `text-base` (16px) on mobile viewports (<768px) to prevent iOS auto-zoom.
- **Thumb Zone Optimization:** Floating GPS Locate Me and bottom sheet handle docked in natural lower thumb reach area.

4. Next.js 16 Component Hierarchy & Boundaries

Component	Runtime	File Location	Architectural Role
<code>LandingMapLayout</code>	SERVER	<code>src/app/page.tsx</code>	Resolves URL query parameters, executes Supabase spatial query, streams listings to client.
<code>MapLibreCanvas</code>	CLIENT	<code>src/components/map/maplibre-canvas.tsx</code>	Initializes WebGL MapLibre GL instance, loads OpenFreeMap vector style, renders price markers.
<code>FloatingSearchCard</code>	CLIENT	<code>src/components/features/floating-search-card.tsx</code>	Desktop 400px floating card, landmark auto-suggest dropdown, collapsible chevron toggle.
<code>MobileBottomSheet</code>	CLIENT	<code>src/components/features/mobile-bottom-sheet.tsx</code>	Mobile 3-snap gesture drawer (Peek/Mid/Full), drag handle, horizontal carousel.
<code>MapOverlayHUD</code>	CLIENT	<code>src/components/map/map-overlay-hud.tsx</code>	Floating "Search this area" button, zoom controls (+/-), GPS Locate Me FAB, Ask AI trigger.
<code>ListingDetailModal</code>	CLIENT	<code>src/components/features/listing-detail-modal.tsx</code>	Full listing detail drawer (hero photo gallery, fee breakdown, jeepney routes 04L/17B, inquiry CTA).

5. Design Tokens & Z-Index Stacking Hierarchy

Layer	Z-Index	Target Elements
Canvas	z-0	WebGL MapLibre base map surface
Vector Overlays	z-10	GIS walking radius circles, jeepney paths
Price Pins	z-20 / z-25	Unselected (z-20) vs Active pin (z-25)
Desktop Panel	z-10	Floating search & results card (400px)
Map Controls	z-30	Zoom buttons (+/-), GPS Locate Me FAB
Search / Pills	z-40	Mobile floating top search pill & filter bar
Bottom Sheet	z-50	Mobile 3-snap bottom sheet drawer
Modals	z-80	Full listing detail modal & photo lightbox

Regional Cebu Palette Compliance:

- **Primary (Maritime Navy):** `#0f2742` — Navigation authority & titles.
- **Action (Tropical Emerald):** `#047857` — Primary CTAs (**5.48:1** contrast).
- **Secondary (Reef Teal):** `#0d9488` — Active filter chips & badges.
- **Background (Sand):** `#faf8f5` — Warm limestone canvas.
- **Warning (Golden Sun):** `#b45309` — Anti-scam GCash warnings (**5.02:1**).

6. Definition of Done & Architectural Sign-Off

- [x] Full-bleed map landing page wireframe designed for both Desktop (1440px) and Mobile (100dvh).
- [x] Desktop 400px floating left card with collapsible chevron and optical map centering specified.
- [x] Mobile 3-snap bottom sheet (Peek 88px, Mid 48dvh, Full 88dvh) and listing detail modal documented.
- [x] Authentic Metro Cebu landmarks (CIT-U, USC, IT Park) and commute jeepney routes (04L, 17B, 13C) integrated.
- [x] WCAG 2.2 AA touch target compliance (44x44px) and 16px mobile input zoom prevention enforced.
- [x] Sprint 1 Boundary Verified: ZERO premature React/JSX feature code created.
- [x] Clean human attribution: **Angel Crushein Yaun (UI/UX Designer)** & **Hermar Centillas (Lead / Scrum Master)**.